Prediction of Drug-Protein Interactions: A Machine Learning Approach

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Abstract

This study investigates the prediction of drug-protein interactions (DPIs) using machine learning. A dataset of epitope sequences, chemical structures, and T-cell receptor data is employed to predict potential interactions. The models explored include Random Forest, XGBoost, and Convolutional Neural Networks (CNN). The results highlight the strengths and weaknesses of each model, with XGBoost being the most effective in terms of accuracy and precision.

1 Introduction

Drug-protein interactions (DPIs) are central to understanding drug efficacy and safety. Traditional methods for identifying these interactions are resource-intensive. This study uses machine learning to automate DPI prediction, improving the speed and efficiency of drug development. By applying advanced algorithms to epitope sequences and chemical structures, we aim to predict interactions more accurately and quickly than traditional experimental methods.

2 Data Preparation

2.1 Dataset

The dataset used comes from the TCR-Epitope Binding Prediction task by the Therapeutic Data Commons. It contains 47,190 entries with attributes:

- Epitope amino acid sequences (epitope_aa)
- Epitope chemical structures (epitope_smi)
- T-cell receptor sequences (tcr)
- Full T-cell receptor sequences (tcr-full)
- Labels indicating interaction (1) or non-interaction (0)

2.2 Data Processing

Tokenization and padding operations were applied to sequence data. Sequences were mapped to integers for computational processing. Feature engineering resulted in a unified feature matrix X of size 47190×600, with labels y.

3 Machine Learning Models

3.1 Random Forest

Random Forest is an ensemble of decision trees that reduces overfitting by averaging predictions from multiple trees. We configured it with 100 trees, max depth of 10, and 5 minimum samples per split. Stratified 5-fold cross-validation was used for performance estimation.

3.2 XGBoost

XGBoost is a gradient boosting model that sequentially builds trees to improve prediction accuracy. We set the max depth to 6 and the learning rate to 0.1. The model is configured for binary classification with logistic regression as the objective function.

3.3 Convolutional Neural Networks (CNN)

CNNs are effective for sequential data, such as protein sequences. We used 4 convolutional layers with increasing filter sizes. Dropout layers were added to prevent overfitting, followed by fully connected layers for enhanced generalization. The output layer used a sigmoid activation for binary classification.

4 Experimental Results

4.1 Model Performance

We evaluated the models based on accuracy, precision, recall, and F1 score. The following are the results:

- **CNN**: Accuracy = 78.25%, Precision = 74.50%, Recall = 85.44%, F1 = 79.60%
- **XGBoost**: Accuracy = 78.34%, Precision = 75.84%, Recall = 77.99%, F1 = 76.90%
- **Random Forest**: Accuracy = 74.09%, Precision = 73.42%, Recall = 75.10%, F1 = 74.25%

4.2 Confusion Matrices

Confusion matrices for each model were analyzed to understand the true positives, false positives, true negatives, and false negatives. XGBoost and CNN showed high precision and recall, with XGBoost slightly outperforming CNN in terms of accuracy. Random Forest had the lowest performance but still demonstrated decent recall.

5 Discussion

XGBoost showed the best overall performance, balancing accuracy and precision. CNN excelled in recall, making it suitable for tasks where detecting positive interactions is critical. Random Forest, while effective in recall, was outperformed by the other models in precision and overall accuracy.

6 Conclusion

This study demonstrates the significant potential of machine learning techniques in predicting drug-protein interactions (DPIs). Among the various models tested, XGBoost proved to be the most effective, combining high accuracy with computational efficiency, making it a promising choice for practical applications. However, while XGBoost has shown excellent results, future work could explore more advanced and complex models, such as deep learning architectures. One such possibility is the use of ProtBERT, a pre-trained model specifically designed to work with protein sequences, which could provide a deeper understanding of protein structure and function. Additionally, other sequence-specific models, designed to handle biological data such as protein sequences, could potentially capture the intricate relationships between drugs and proteins more accurately. By leveraging deep learning models or pre-trained models specialized for proteins, the prediction performance for DPIs could be further improved, leading to more reliable and actionable insights. Moreover, integrating these models with larger and more diverse datasets could further enhance the generalizability and robustness of the predictions.

7 References

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